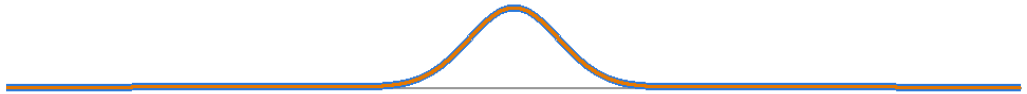
4.10 Wave Interference

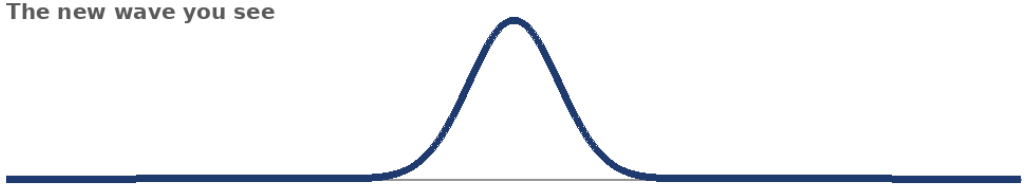
What happens when two waves meet?

CONSTRUCTIVE: crest meets crest

Wave A (orange) and Wave B (blue)



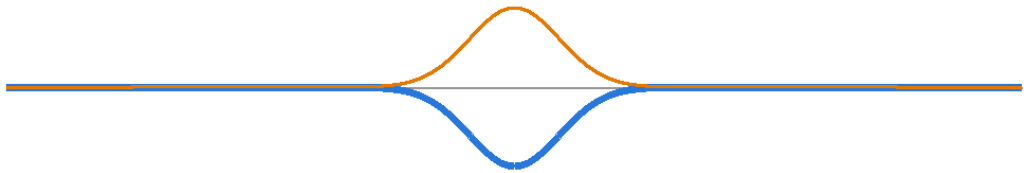
The new wave you see



BIGGER: 3 + 3 = 6

DESTRUCTIVE: crest meets trough

Wave A (orange) and Wave B (blue)



The new wave you see

FLAT: 3 + (−3) = 0

Think About It

You drop two pebbles into a calm pool. The rings of ripples spread out and meet. What happens?

Single choice poll

A. They bounce off each other

Single choice poll

B. They pass through each other and combine for a moment

Single choice poll

C. They stop and disappear

Single choice poll

D. They stick together forever

Today's Goals

I can explain what happens when two waves meet.

I can tell constructive from destructive interference.

I can add amplitudes to find the new wave.

I can give real-world examples of interference.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Waves can be described by their speed, wavelength, amplitude, and frequency.
- Vibrations cause wavelike disturbances that transfer energy from one place to another.
- Today: when waves overlap, their amplitudes combine.

In this lesson

- Integers: Power Up! + Case #1
- Wave interference video
- Constructive vs. destructive
- Add the amplitudes
- Real-world interference
- Part 2: Interference Missions

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

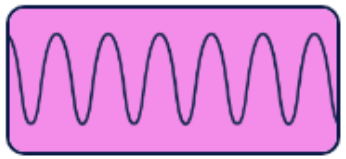
Science 7 · Week 4: Nov 30 – Dec 4

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · NOV 30
4.10: Wave Interference

- **WARM-UP** Integers: **Power Up!** + **Case #1** (ripples).
- *READ* Read lesson **4.10: Wave Interference**.
- **CLASSKICK** Complete the **4.10** Classkick slides: Part 1 and the **Interference Missions** game.
- **DUE!** **4.07 Quest** is due today.
- **EXT DUE!** Extended time: **4.03 Quest** and **4.04 Refraction Lab** close today.



TUESDAY · DEC 1
4.11: Sound Waves Lab

- **WARM-UP** Integers: **Case #2** (decibels).
- *READ* Read lesson **4.11: Sound Waves Lab**.
- **CLASSKICK** Complete the **4.11** Classkick slides: Parts 1–2 (PhET Sound Waves).



WEDNESDAY · DEC 2
4.12: Explore Sound in a Medium

- **WARM-UP** Integers: **Case #3** (lightning).
- *READ* Read lesson **4.12: Explore Sound in a Medium**.
- **CLASSKICK** Complete the **4.12** Classkick slides: Part 1 and the **Wave Speed Lab**.



THURSDAY · DEC 3
4.13: Unit 4 Review

- **WARM-UP** Integers: **Case #4** (the seiche).
- **CLASSKICK** **4.13 Review** slide, the **Vocab Review** (2 slides), and the **4.14 Mission Map** (we do it together in class).
- **REVIEW** Play the **Unit 4 Wordwall** review game on the 4.13 page.
- **SURVEY** Take the **Unit 4 vocabulary post test**. Closes Thu, Dec 10.

FRIDAY · DEC 4
Science Friday

- **FINISH** Finish any lessons, Classkick slides, or assignments you have not completed this week.
- **REDO** Check your grades. Lower than a B? Review the Classkick notes and redo it.
- **CLASSKICK** Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.
- **STUDY** Review your Mission Map and Unit 4 slides for Monday's test.

INTEGER CREW: POWER UP!

Multiplying Integers

The crew's power gate is locked! Match the signs to choose the charge.

SAME signs? POSITIVE power! | DIFFERENT signs? NEGATIVE power!

SAME SIGNS → POSITIVE

Multiply the counts. Make it positive.



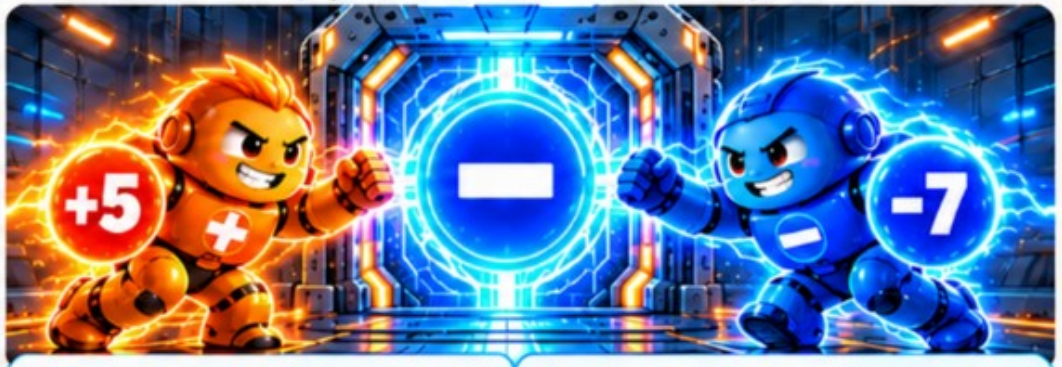
$(-6) \times (-2) = 12$



$6 \times 2 = 12$

DIFFERENT SIGNS → NEGATIVE

Multiply the counts. Make it negative.



$5 \times (-7) = -35$

$(-5) \times 7 = -35$

REPEATED ADDITION: $3 \times (-4) = (-4) + (-4) + (-4) = -12$

Any number × 0 = 0. Robots represent signed numbers.

CASE #1 • MONDAY

Is the trough deeper than −5 cm?

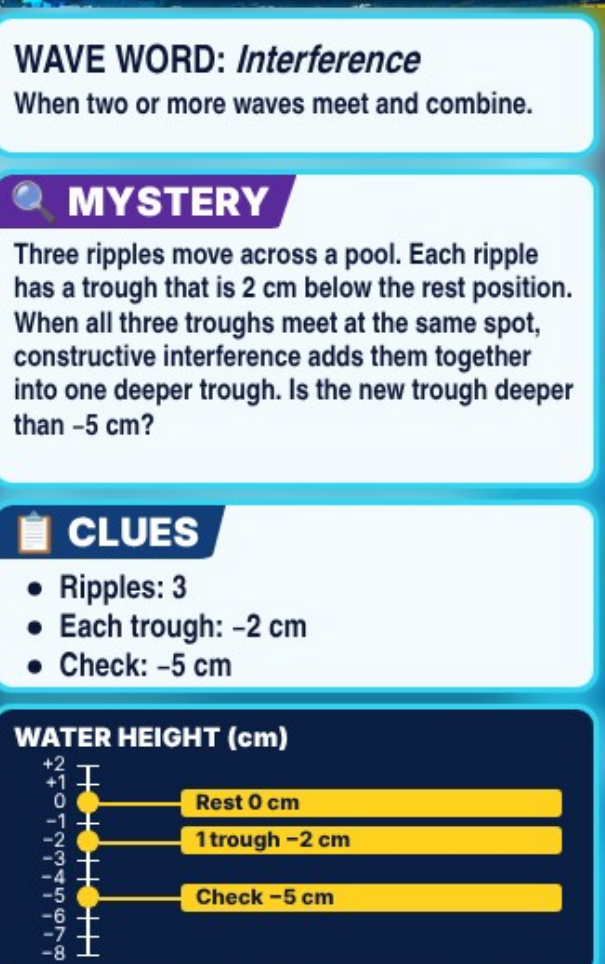
WAVE WORD: *Interference*
When two or more waves meet and combine.

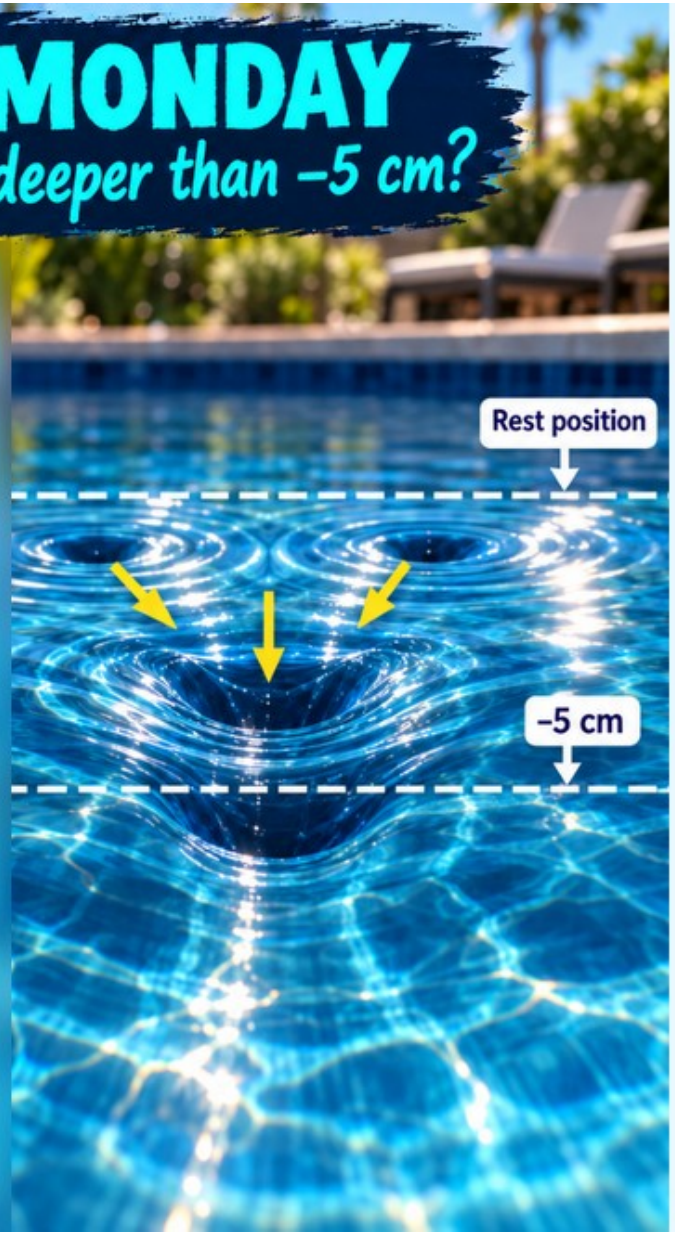
MYSTERY

Three ripples move across a pool. Each ripple has a trough that is 2 cm below the rest position. When all three troughs meet at the same spot, constructive interference adds them together into one deeper trough. Is the new trough deeper than −5 cm?

CLUES

- Ripples: 3
- Each trough: −2 cm
- Check: −5 cm





CLAIM

Is the new trough deeper than −5 cm?

☐ YES ☐ NO

EVIDENCE

Solve:

$3 \times (-2) =$ _____

Repeated addition: (−2) + (−2) + (−2) = _____

REASONING

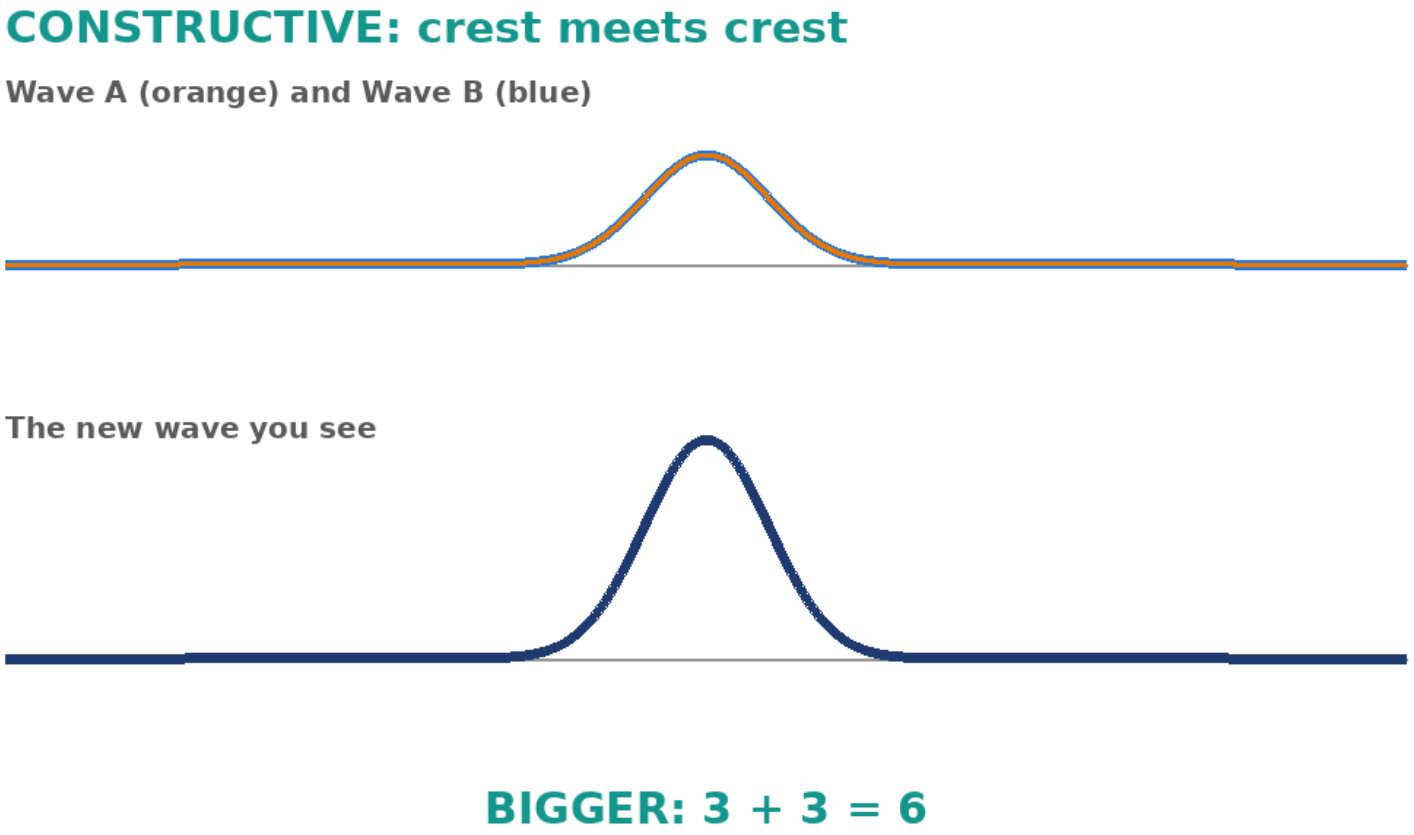
Finish the sentence:

The signs are _____, so the answer is _____. The new trough is at _____ cm. −6 cm is _____ than −5 cm.

Watch: Wave Interference



Constructive Interference

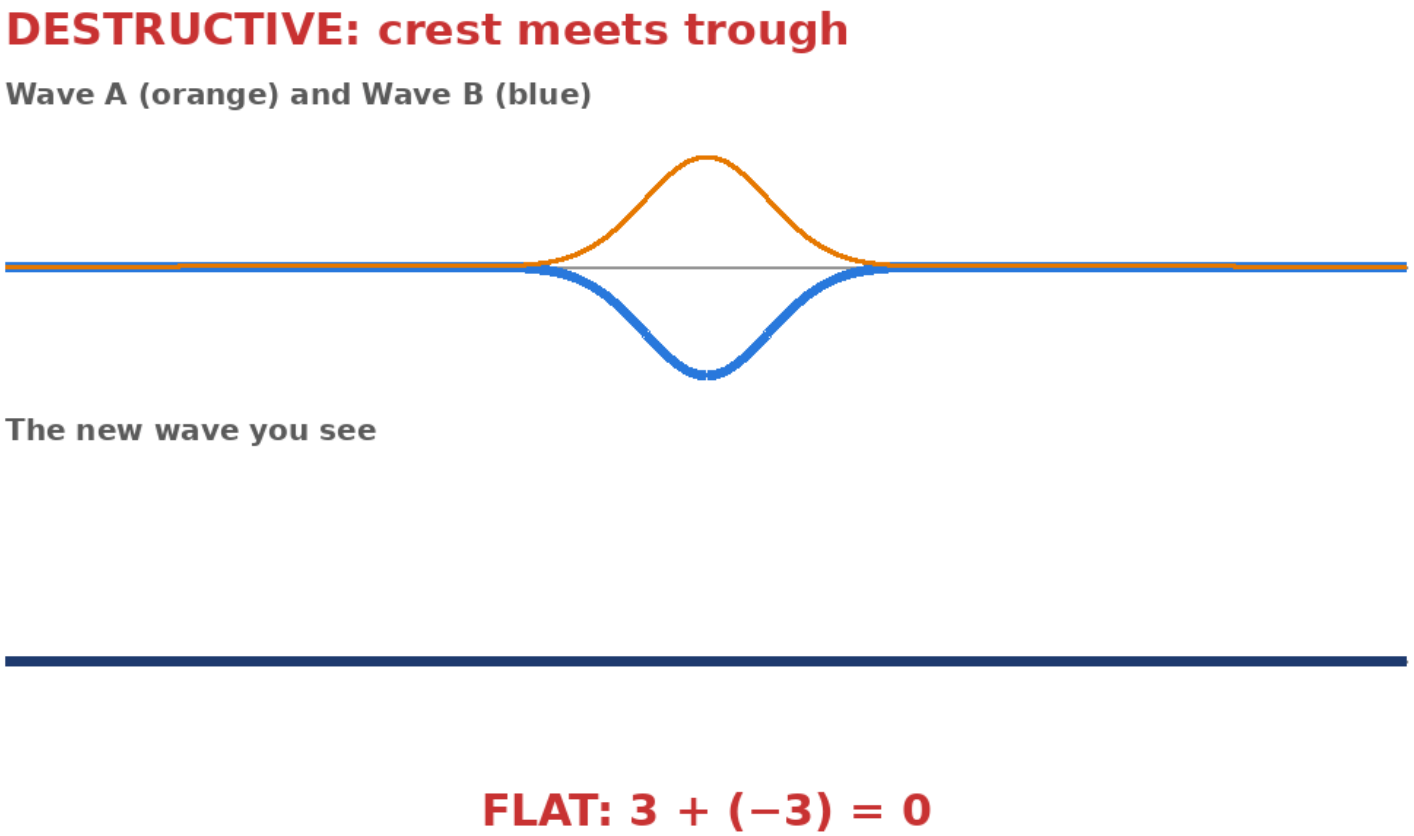


Crest meets crest (in phase).

The amplitudes ADD. The new wave is bigger.

Like two friends pushing a swing at the same time: it goes higher.

Destructive Interference

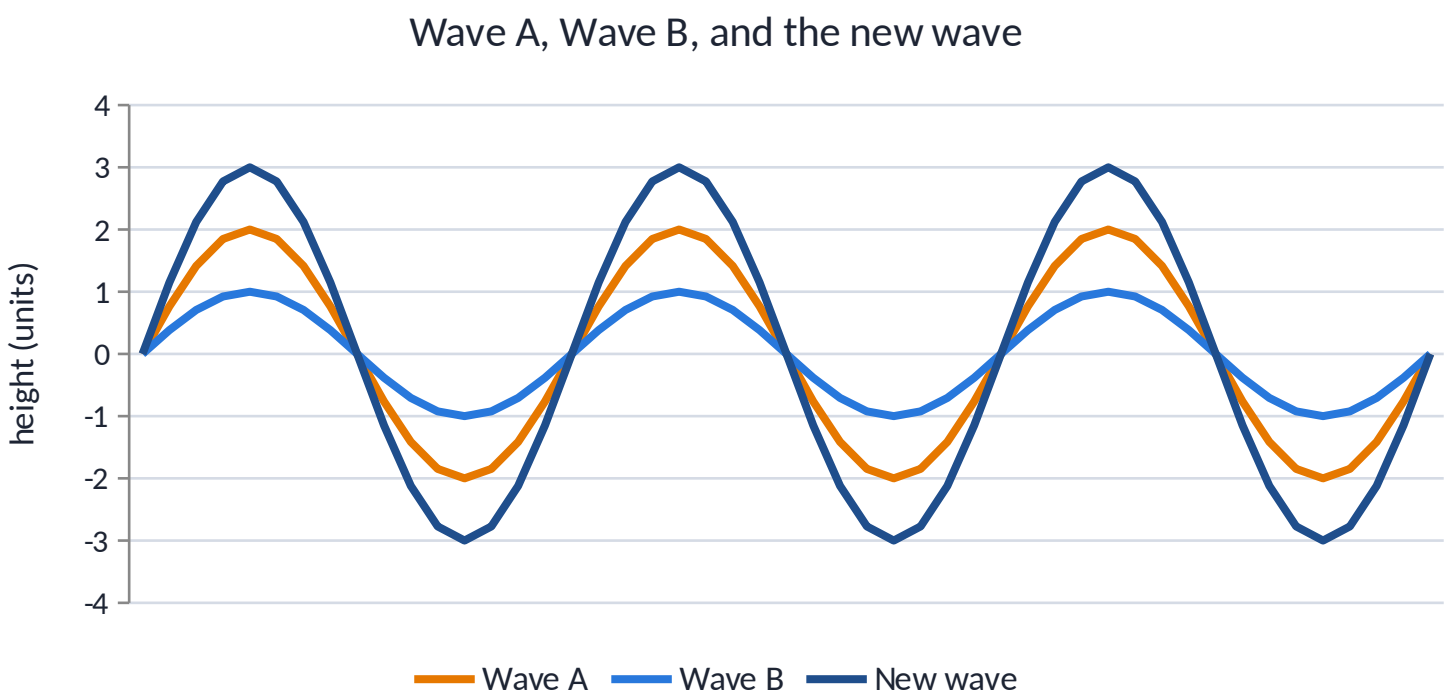


Crest meets trough (out of phase).

The new wave is smaller. Same size? It cancels out to zero.

After they meet, both waves keep going on their own paths, unchanged.

Graph Poll



Read the graph.

Wave A and Wave B meet and make the navy wave. What kind of interference is this?

Single choice poll

A. Constructive

Single choice poll

B. Destructive

Single choice poll

C. Refraction

Single choice poll

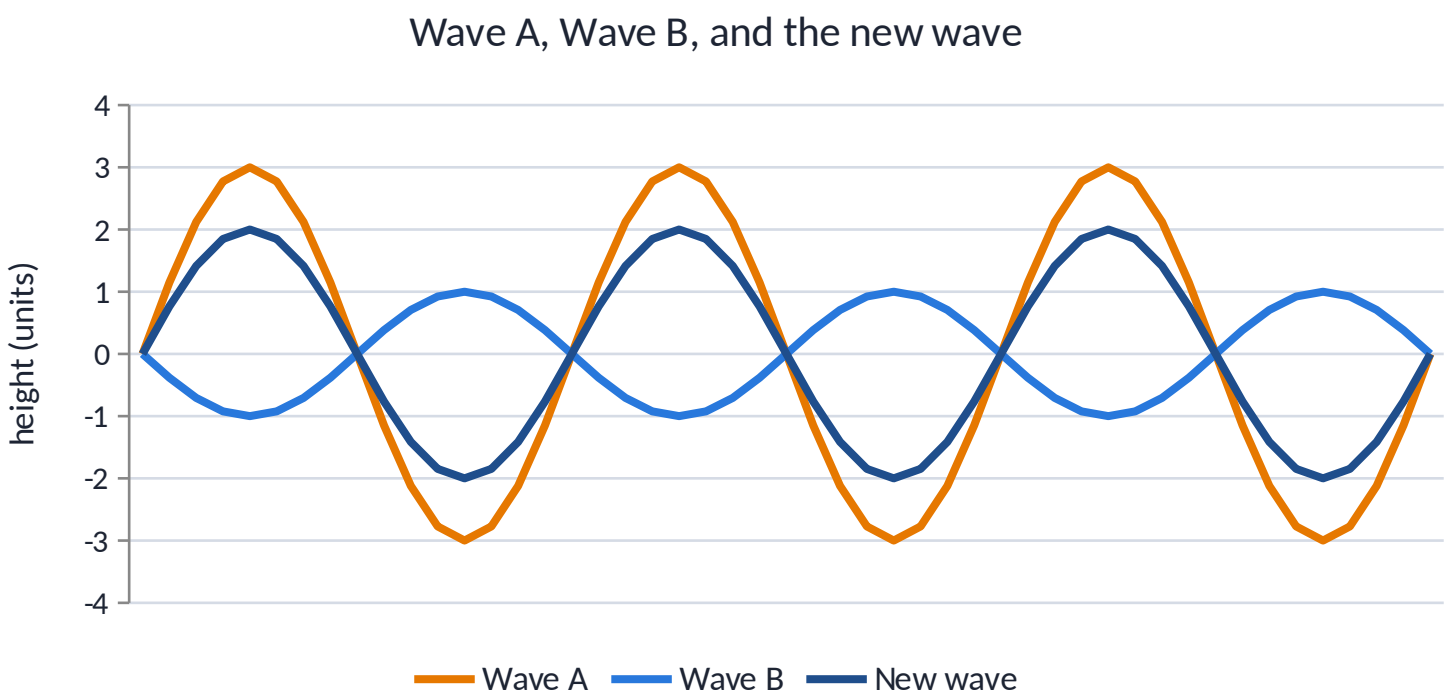
D. Reflection

Add the Amplitudes (Use Integers!)

Wave A	Wave B	Add them	New wave
crest +1	crest +3	1 + 3	+4 (bigger)
crest +4	trough −1	4 + (−1)	+3 (smaller)
trough −2	trough −2	(−2) + (−2)	−4 (deeper trough)
crest +5	trough −5	5 + (−5)	0 (flat!)

Crest = positive (+). Trough = negative (−). Add them like integers to get the new wave.

Graph Poll



Read the graph.

Wave A (3 units) meets Wave B, crest to trough. What is the amplitude of the new wave?

Single choice poll

A. 4 units

Single choice poll

B. 2 units

Single choice poll

C. 3 units

Single choice poll

D. 0 units

Interference in Real Life

Noise-canceling headphones

They make a sound wave that is the opposite of the noise. Crest meets trough: quiet!

Destructive

Surfing waves

Two ocean swells line up crest to crest and make a bigger wave.

Constructive

Radio dead spots

Waves from two towers or stations cancel in some spots, so reception is poor.

Destructive

Soap bubbles

Light waves reflecting off the bubble combine and make rainbow colors.

Both

Engageli Poll

Two kids jump on a trampoline at exactly the same moment, and they both bounce higher. What kind of interference is this most like?

Single choice poll

A. Constructive

Single choice poll

B. Destructive

Single choice poll

C. Refraction

Single choice poll

D. No interference

Who Is Right?

A crest and a trough of the same size meet and the rope goes flat. What happens to the two waves next?

Ava says:

"They are gone forever, because they canceled each other out."

Ben says:

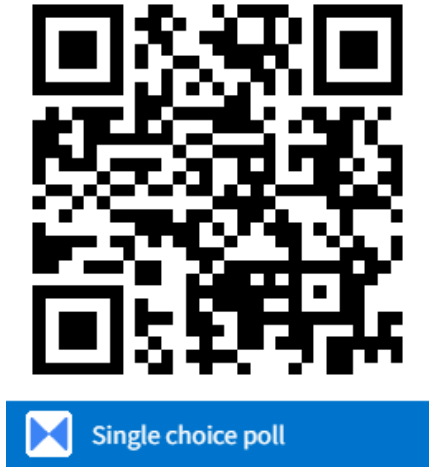
"They bounce off each other and go back the way they came."

Cam says:

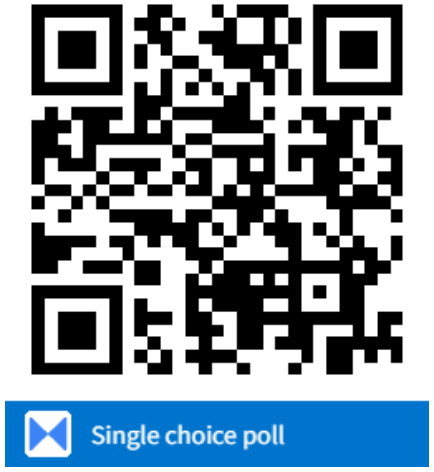
"They pass through each other and keep going, unchanged."

Dee says:

"They stick together and make one flat wave forever."



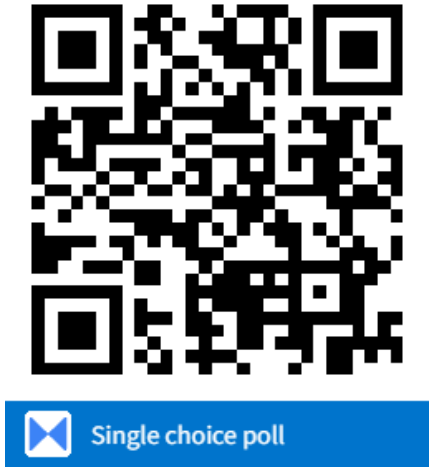
A. Ava



B. Ben



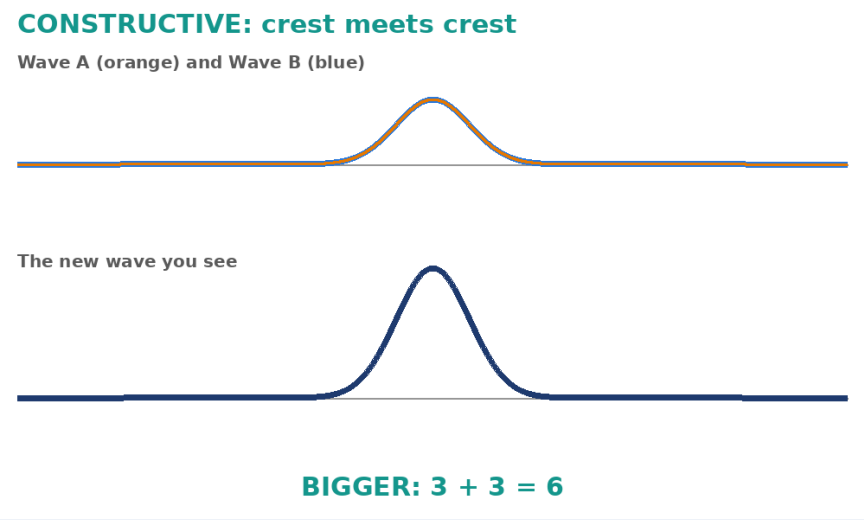
C. Cam



D. Dee

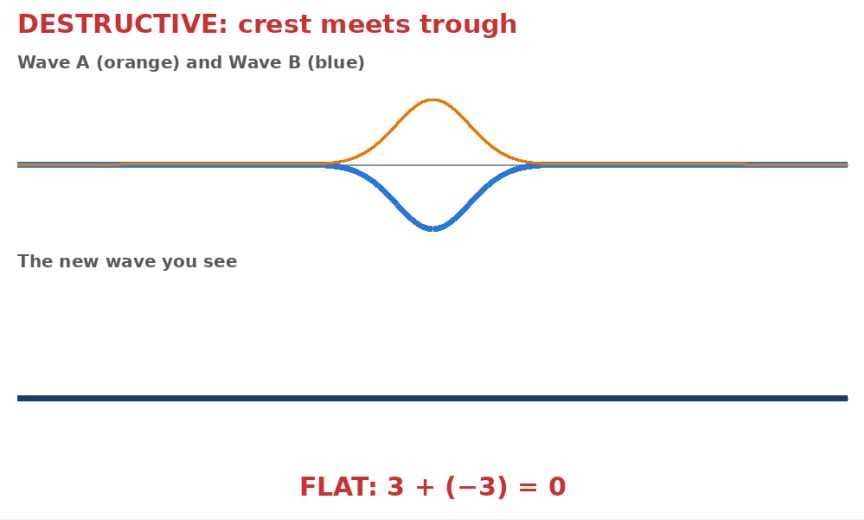
Compare: Constructive vs. Destructive

Constructive



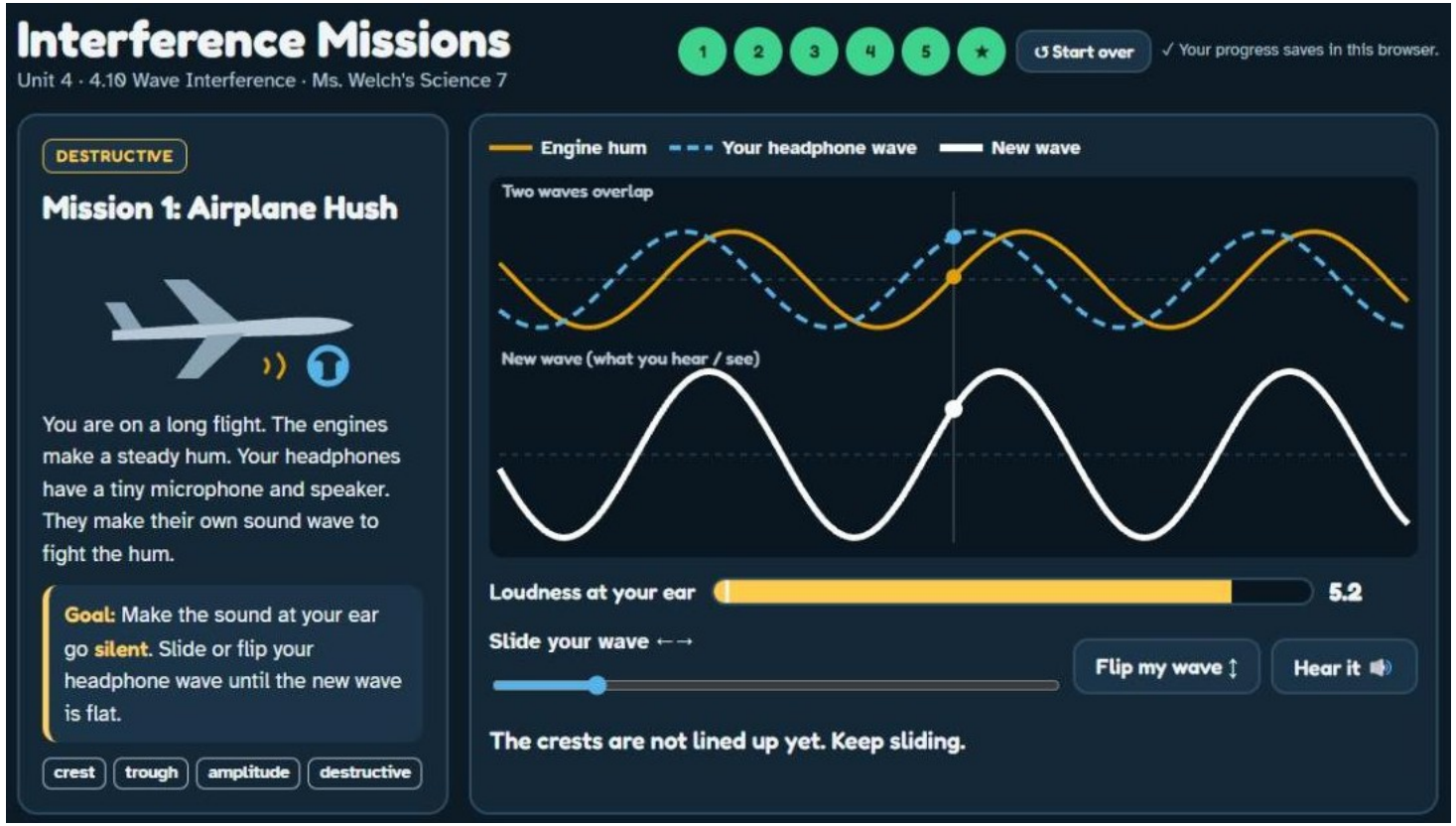
- crest meets crest (in phase)
- amplitudes add
- bigger wave

Destructive



- crest meets trough (out of phase)
- amplitudes subtract
- smaller wave, or flat (0)

Part 2: Interference Missions



▶️ Open Interference Missions

Use the rest of class

1. Airplane Hush
2. Surf's Up
3. Harbor Engineer
4. Lawn Mower Level
5. Road Trip Radio
6. Lightning Round

Check off each mission on your Classkick slide. Your progress saves in this browser.

Finish: Code and Certificate

1. Type your code

Finish all 6 missions. Type the completion code in the Q7 box.

2. Copy the certificate

Click Copy certificate in the game (or snip it: Windows + Shift + S).

3. Paste it

Click the box on the Classkick slide and press Ctrl + V.

Key Ideas

1. Interference: two waves overlap and combine into a new wave.
2. Constructive: crest meets crest. Amplitudes add. Bigger wave.
3. Destructive: crest meets trough. Smaller wave, or flat if they are the same size.
4. After they meet, the waves keep going, unchanged.

Re-rate your goals. Not done? Finish Interference Missions and paste your certificate.